

Problem 15

Evaluate the integral:

$$\int \frac{e^{3x} + 1}{e^x + 1} dx.$$

Solution

We are tasked with solving the integral:

$$\int \frac{e^{3x} + 1}{e^x + 1} dx.$$

Step 1: Simplify the integrand

Let $u = e^x$. Then, $du = e^x dx$, which implies $dx = \frac{du}{u}$. Additionally, note that:

$$e^{3x} = (e^x)^3 = u^3.$$

Substitute $e^x = u$ into the integral:

$$\int \frac{e^{3x} + 1}{e^x + 1} dx = \int \frac{u^3 + 1}{u + 1} \cdot \frac{du}{u}.$$

Simplify the fraction $\frac{u^3+1}{u+1}$. Using polynomial division, write:

$$u^3 + 1 = (u + 1)(u^2 - u + 1).$$

Thus:

$$\frac{u^3 + 1}{u + 1} = u^2 - u + 1.$$

Substitute this into the integral:

$$\int \frac{e^{3x} + 1}{e^x + 1} dx = \int \frac{u^2 - u + 1}{u} du.$$

Step 2: Simplify the integrand further

Break the fraction into separate terms:

$$\frac{u^2 - u + 1}{u} = u - 1 + \frac{1}{u}.$$

Thus, the integral becomes:

$$\int \frac{e^{3x} + 1}{e^x + 1} dx = \int \left(u - 1 + \frac{1}{u} \right) du.$$

Step 3: Evaluate the integral

Now, integrate each term:

$$\int \left(u - 1 + \frac{1}{u} \right) du = \int u du - \int 1 du + \int \frac{1}{u} du.$$

1. The first term:

$$\int u du = \frac{u^2}{2}.$$

2. The second term:

$$\int 1 \, du = u.$$

3. The third term:

$$\int \frac{1}{u} \, du = \ln |u|.$$

Combine the results:

$$\int \left(u - 1 + \frac{1}{u} \right) \, du = \frac{u^2}{2} - u + \ln |u| + C,$$

where C is the constant of integration.

Step 4: Substitute back $u = e^x$

Recall that $u = e^x$. Substitute $u = e^x$ back into the result:

$$\int \frac{e^{3x} + 1}{e^x + 1} \, dx = \frac{(e^x)^2}{2} - e^x + \ln |e^x| + C.$$

Simplify $\ln |e^x|$ to x :

$$\int \frac{e^{3x} + 1}{e^x + 1} \, dx = \frac{e^{2x}}{2} - e^x + x + C.$$

Final Answer

The solution to the integral is:

$$\int \frac{e^{3x} + 1}{e^x + 1} \, dx = \frac{e^{2x}}{2} - e^x + x + C.$$